LESSON 3.4
USING OPERATIONS TO SOLVE REAL-WORLD PROBLEMS – PART 1

LESSON INTRODUCTION

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

LEARNING TARGETS

- I can determine which operation should be used to solve a real-world problem.

BENCHMARK COHERENCE

BUILDING ON

MA.5.AR.1 Solve problems involving the four operations with whole numbers and fractions

WORKING TOWARD

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- What information do you need to determine the mathematical operation required to create a numeric expression?
- How could you use a mathematical operation to represent a real-world scenario?

LESSON NARRATIVE

In this lesson, students develop their ability to interpret word problems and determine the operations needed to solve these problems. Using real-world problems with positive decimals and fractions, including mixed numbers, students explore how to sketch or explain their reasoning and create numeric expressions. Students then use developed problem-solving strategies to identify which operations to use to solve real-world problems in the Guided Practice. The Wrap-Up provides students with an opportunity to reflect on learning and to rate their confidence in interpreting which operation should be used for a real-world problem.

COMPONENT SUMMARY

3.4.1 WARM-UP *(~5 MIN.)*

Error analysis on discount prices

3.4.2 GUIDED INSTRUCTION *(20 – 25 MIN.)*

Real-world scenarios to determine the appropriate operations for which to write expressions

3.4.3 GUIDED PRACTICE *(20 – 25 MIN.)*

Practice identifying the appropriate operations required to solve real-world problems

3.4.4 WRAP-UP *(~5 MIN.)*

Students determine appropriate operation and rate confidence in their answers

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Numeric expression
- Operation

MATERIALS

- (Optional) Chart paper
- (Optional) Highlighters
- (Optional) Markers

ADDITIONAL PREPARATION

- (Optional)If creating the class anchor chart for 3.4.2 Guided Instruction, label chart paper with one operation each at the top prior to the lesson. Gather markers for students to fill in the chart paper with common key words/phrases used to determine the required operation.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may use the incorrect operation when translating real-world scenarios into numeric expressions.
- Students may create inaccurate numeric expressions.
- Students may write square inches for the units when evaluating the problems involving inches.

THINGS TO CONSIDER

- Position students to recall numeric expressions by having them draw a visual model first and then create their numeric expressions from the visual model.
- Students should have experience performing operations on problems involving multi-digit decimals and fractions. Consider opportunities for brief algorithm reviews during the lesson if needed.

LITERACY AND LANGUAGE ROUTINES

Contemplate, Then Calculate

Lessons 4-9 in this unit emphasize real-world contexts. Utilize Lesson 4 as a training ground for the process rather than the product of solving, highlighting, and recognizing students' thinking processes rather than their answers. 3.4.3 Guided Practice provides a great opportunity for this using the Contemplate, Then Calculate routine.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Consider highlighting clarification language in MA.K12.MTR.7.1 that can support the analysis and strategies for problem-solving.

- Connect…to everyday experiences
- Redesign models or methods to improve accuracy…
- …validate conclusions by comparing…
- …question the accuracy of models / methods…

DIFFERENTIATE INSTRUCTION

Provide students with additional scaffolds including the "Cover-Up" method or replacing the fractions and decimals with whole numbers, allowing them to focus on understanding the problem-solving process and how to interpret which operation should be used, rather than the computation.

During 3.4.3 Guided Practice, support auditory learners by using a Think Aloud as you analyze the scenarios for which operation is used. This practice may encourage metacognition as students learn about different ways that others talk themselves through problem-solving strategies.

3.4.1 WARM-UP: MATH FLOP

Component Overview: *Consider having students Think-Pair-Share as they analyze the sales tags and notice errors. As students share to the whole group, students can show whether they agree or disagree to engage in discussion as the errors are identified and fixed.*

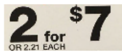
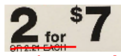
TEACHER MOVES

Think-Pair-Share

Consider having students first individually think about and work on the columns in the table as they analyze the errors, then take 1 to 2 minutes to share their responses or progress with a partner. As partners share, listen for strategies or ideas that accurately identify the mistakes, then select these partners to share their thoughts with the class.

3.4.1 Warm-Up: Math Flop

1. **Determine the mistake in each picture. Then, complete the table.**

Math Flop	**The Mistake:** Which number is not correct?	**My Work:** Show the math that proves it is a mistake.	**The Correct Label:** Fix the label by marking out the mistake and inserting the correct number.
	*Save $1.49*	$\begin{array}{r} 3.99 \\ -2.99 \\ \hline 1.00 \end{array}$	 *Save $1.00*
	*Or $2.21 each*	$\frac{7}{2} = 3.50$	 *Or $3.50 each*

3.4.2 Guided Instruction: Interpreting Real-World Problems

1. Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in.
Read each question and complete the following table.
   a. Which mathematical operation can be used to find the total length of the two boards?

Operation	Sketch a picture or explain why you chose that operation.
Addition	*Answers will vary. The word total is a key word that means to add. When you are finding a total, you combine the two amounts, and to combine two totals, you add.*

   b. Which mathematical operation can be used to find the length of each piece when the longest board is cut into four equal-sized pieces?

Operation	Sketch a picture or explain why you chose that operation.
Division	*Answers will vary. The word each is a key word that means division. The problem also states that the board needs to be in 4 equal-sized pieces which indicates the board should be divided by 4.*

   c. Check your answers with your partner and resolve any differences if necessary.

   d. Which mathematical operation can be used to determine the difference in the board lengths?
   Subtraction

   e. Which mathematical operation can be used to find the total length of 48 of the shortest boards lined up end-to-end?
   Multiplication

3.4.2 GUIDED INSTRUCTION: INTERPRETING REAL-WORLD PROBLEMS

Component Overview: *Work as a class through questions 1a-1e by drawing the pictures and writing the operations together. Discussions should focus on how students know which operation to use. The first section focuses on identifying and justifying the required operation only, then the second section progresses to creating numeric expressions given real-world expressions. Have students work individually or in partners to write the numerical expressions as they complete the table in question 1f, while circulating the room to check work for accuracy and answering questions.*

TEACHER MOVES

Avoid teaching key words. This is an unreliable trick that should be avoided. Instead, have students use the strategies built in the lessons to think through the process. Sketching pictures or verbalizing a context would be helpful.

ENRICHMENT

Does your classroom utilize a specific problem-solving strategy? How do your students incorporate MA.K12.MTR.7.1 strategies in practice for real-world contexts? Consider modeling a Think Aloud for question 1a using prompts that reflect your classroom strategies.

- What is the question asking me to do?
- Can I make a sketch of the context?
- Can I restate the problem and my approach to solving it in my own words?
- Can I explain the context to a 5th grader?
- Does this answer seem reasonable?

3.4.2 GUIDED INSTRUCTION: INTERPRETING REAL-WORLD PROBLEMS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Operations with fractions may be a barrier to learning in this activity.

Consider having students cover up the value to draw a picture then label the picture. Another possibility is to replace fractions with whole numbers for a lower entry point to remove barriers from certain groups of students and then re-establishing the context with fractions.

TEACHER MOVES

Notice that students are not actually calculating the solution in order to center the focus of the teaching and learning on the interpretation and approach to solving.

QUESTIONING

If students are working in small groups or on their own for the final portion of the table, consider challenging some students to actually find the solution for Louis.

f. **Write the numeric expression that shows how to calculate the answer. Then, determine the units of the answer.**

Information	Numeric Expression	Units of the Answer
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the total length of the two boards?	$27\frac{3}{8} + 14\frac{5}{12}$	*inches*
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the length of each piece when the longest board is cut into four equal-sized pieces?	$27\frac{3}{8} \div 4$	*inches*
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the difference between the board lengths?	$27\frac{3}{8} - 14\frac{5}{12}$	*inches*
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the total length of 48 of the shortest boards lined up end-to-end?	$48 \cdot 14\frac{5}{12}$	*inches*

3.4.3 Guided Practice: Operations in Real-World Context

1. The table contains four different real-world contexts. Select the question in the middle column that requires the use of the operation indicated in the last column within this context.

Real-World Context	Which Question?	Operation
A Valencia orange tree that is twelve years old can produce 1.4 boxes of oranges. In May 2017, a box of fresh Valencia oranges cost \$20.45.	o For how many years did a box of Valencia oranges cost \$20.45? o How much money would a 12-year old Valencia orange tree generate? o What is the price per box of Valencia oranges?	$\times$
In 2016, there were 19.3 million Snapchat users in the age range of 18-24, and 16.7 million users in the age range of 25-34.	o How many Snapchat users were there between the age of 18 and 34? o What was the difference between the number of Snapchat users in the 18-24 and the 25-34 age range? o How many Snapchat users were younger than 18?	$+$
In 2004, each person in the United States consumed 61.5 pounds of refined sugar on average. In 2008, the average consumed per person was 66.3 pounds of refined sugar.	o What is the total amount of sugar the average person consumed in 2004 and 2008? o What was the average yearly increase in sugar consumption from 2004 to 2008? o What is the difference between the amount of sugar consumed in 2004 and 2008?	$-$
Barbara sent letters that exceeded the allowed weight by the U.S. Postal Service. To mail the letters she had to add a \$0.15 stamp to each letter. She spent \$2.55 in additional weight charges.	o What was the total cost of sending the letters? o For how many letters did Barbara pay the additional weight charge? o What is the difference between the cost of sending a regular letter and one that exceeds the weight allowed?	$\div$

3.4.3 GUIDED PRACTICE: OPERATIONS IN REAL-WORLD CONTEXT

Component Overview: *Students identify which question uses the correct operation to describe a real-world context.*

TEACHER MOVES

Contemplate, Then Calculate

Constructing an approach or strategy before solving may not come naturally to many students who are accustomed to “just give me the answer” math. Consider approaching this activity with the “Contemplate, Then Calculate” language routine that encourages students to think before doing. Instead of having students actually calculate, the student should state what operation is needed to calculate.

ENRICHMENT

Think-Aloud Sample: Snapchat

- Which question would require me to add something?
- What is the information in the question?
- If there are 19.3 million in one range and 16.7 million in the other range, maybe ask a question about both ranges together or total.
- “How many users between 18 and 34?” This is both ranges total, so yes, I would have to add to find both ranges.

DIFFERENTIATE INSTRUCTION

If utilizing small-group learning, consider having students take turns modeling a Think Aloud for each other, illuminating different ways of thinking and entry points for different types of learners.

3.4.3 GUIDED PRACTICE: OPERATIONS IN REAL-WORLD CONTEXT (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider putting students into groups of four. Each student has their own problem to solve and should share their thinking with the other students in their group. The teacher then can call on different groups for the person that had the problem to share their thinking with the whole class.

QUESTIONING

To extend this activity, students who finish early can create their own real-world contexts, numeric expressions, and answers with correct units to add to the table. Consider asking these students to share their work with a partner to check for accuracy.

2. Use the same four real-world contexts and the operation indicated to find the answer to the question selected in the previous table. Write the numeric expression used to determine the answer. Then, show your work and write the answer using the correct units.

Information	Numeric expression	Show your work and answer with units
A Valencia orange tree that is 12 years old can produce 1.4 boxes of oranges. In May 2017, a box of fresh Valencia oranges cost \$20.45.	$20.45 \cdot 1.4$	\$28.63
In 2016, there were 19.3 million Snapchat users in the age range of 18-24 and 16.7 million users in the age range of 25-34.	$19.3 + 16.7$	36 *million*
In 2004, each person in the United States consumed 61.5 pounds of refined sugar on average. In 2008, the average consumed per person was 66.3 pounds of refined sugar.	$66.3 - 61.5$	4.8 *pounds*
Barbara sent letters that exceeded the allowed weight by the U.S. Postal Service. To mail the letters, she had to add a \$0.15 stamp to each letter. She spent \$2.55 in additional weight charges.	$2.55 \div 0.15$	17 *letters*

3.4.4 Wrap-Up: Reflection

Mathematics is most useful in the real world. Determine the operation(s) needed to find the answer to each question. Then, mark how confident you are in your choice of operation(s) to be used.

1. **Fitzgerald is selling chocolate bars to raise money for a trip. He has to sell a total of 4 boxes. The first day, he sold $2\frac{3}{4}$ boxes. How many more boxes does he have to sell?**
   The operation(s) that should be used is/are ***subtraction*****.**

0 1 2 3 4 5 6 7 8 9 10

Answers will vary.

2. **Ginger is buying new carpet for her living room. The carpet costs \$56.81 per square yard with an additional \$7.64 per square yard to install it. The size of the living room is 13.6 square yards. How much will the carpet for the living room cost?**
   The operation(s) that should be used is/are ***multiplication and addition*****.**

0 1 2 3 4 5 6 7 8 9 10

Answers will vary.

3.4.4 WRAP-UP: REFLECTION

Component Overview: *As an assessment of learning targets, students individually determine the operation needed for given real-world scenarios, then rate their confidence in the accuracy of their answer choices. Data and observations from the lesson and the Wrap-Up can be used to determine where students stand for the next lessons in this unit. Students should not use a calculator to complete the Wrap-Up.*

DIFFERENTIATE INSTRUCTION

Considering the variety of correct or incorrect responses for each prompt and what those responses are, along with the emoji scale, can tell you about learning progression.

Question 1: Students could choose addition if they estimate $1\frac{1}{4}$ to then add to $2\frac{3}{4}$. Both addition or subtraction could be correct and will tell you how that student is approaching problem-solving.

Question 2: Similar to question 1, both multiplication and division can be considered accurate depending on the students' approach. Use the emoji scale to better determine whether the student was guessing or finding a creative entry point.

